

Predictive Modeling of White Wine Quality Using Machine Learning Approaches

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Abstract

Wine quality assessment traditionally relies on human sensory evaluation, which is subjective and time-consuming. With advances in machine learning, data-driven models can assist in predicting wine quality more objectively. This study applies exploratory data analysis and a Random Forest classifier to the *Wine Quality (White)* dataset to develop a predictive model using physicochemical features. Correlation heatmaps, scatterplot matrices, and violin plots were utilized for pattern exploration. The Random Forest model was trained using 80% of the dataset and tested on the remaining 20%. The model achieved a classification accuracy of approximately 69%, with alcohol, residual sugar, and density identified as the most influential predictors. The results demonstrate that machine learning can effectively classify wine quality and may support automated quality assessment systems.

I. Background

Wine quality evaluation is traditionally performed by trained experts, which introduces variability and operational costs. With the increasing availability of wine production data, machine learning provides a promising alternative for objective prediction. The UCI Wine Quality dataset contains chemical measurements of white wines, offering an opportunity to explore how physicochemical features relate to sensory-based quality scoring.

Previous studies have shown that models such as Random Forests, Gradient Boosting, and Support Vector Machines can predict wine quality with reasonable accuracy. However, the classification remains challenging due to class imbalance and overlapping feature patterns. This study extends this body of work by applying exploratory data visualization and model training

to evaluate how physicochemical variables influence wine quality.

II. Objectives

The aim of this study is to develop a predictive model capable of classifying wine quality using physicochemical properties. The specific objectives are:

1. To explore and visualize relationships between chemical composition and wine quality.
2. To train a machine learning model for wine quality classification.
3. To evaluate prediction performance using accuracy, precision, recall, and F1-score.
4. To identify key predictors contributing to model performance.

III. Methodology

3.1 Dataset

The dataset used in this study was the *White Wine Quality* dataset, consisting of chemical analysis and quality scores. The dataset contained multiple variables including fixed acidity, volatile acidity, citric acid, residual sugar, chlorides, sulfur dioxide levels, density, pH, sulphates, alcohol content, and a quality score ranging from 0–10.

3.2 Exploratory Data Analysis

Correlation heatmaps, scatterplot matrices, and violin plots were generated using ggplot2, reshape2, and GGally. These visualizations provided insights into patterns, distributions, and potential predictive relationships.

3.3 Model Development

The quality variable was converted to a factor to support classification. The dataset was randomly split into training (80%) and testing (20%) subsets. A Random Forest classification model was developed using:

- **ntree = 500**
- **mtry = 4**

3.4 Evaluation Metrics

Model performance was assessed using:

- Confusion matrix
- Overall accuracy

$$Accuracy = \frac{TP + TN}{FP + FN + TP + TN} = \frac{\sum_{i=1}^k cm_{ii}}{\sum_{i=1}^k \sum_{j=1}^k cm_{ij}}$$

cm = confusion matrix

k = Total number of classes in the classification problem

$\sum_{i=1}^k cm_{ii}$ = sum of diagonal elements

$\sum_{i=1}^k \sum_{j=1}^k cm_{ij}$ = Total samples in the dataset

- Precision, recall, and F1-score per class

$$Precision_i = \frac{TP}{TP + FP} = \frac{cm_{ii}}{\sum_{j=1}^k cm_{ij}}$$

cm_{ii} = value on the diagonal

$\sum_{j=1}^k cm_{ij}$ = total number of samples predicted to belong to class i

$\sum_{j=1}^k cm_{ji}$ = total number of samples that actually belong to class i

3.5 Model Selection Rationale

Although the wine quality variable is stored on a numerical scale from 0–10, the meaning of this value is not continuous, but rather categorical and ordinal. Each value represents a discrete tasting score assigned by human evaluators, not a precise measured quantity. Therefore, the difference between scores is qualitative rather than mathematically proportional (e.g., a wine rated 6 is not “twice as good” as one rated 3). Treating such values as continuous in a regression model, risks imposing artificial numeric assumptions that the data do not inherently support.

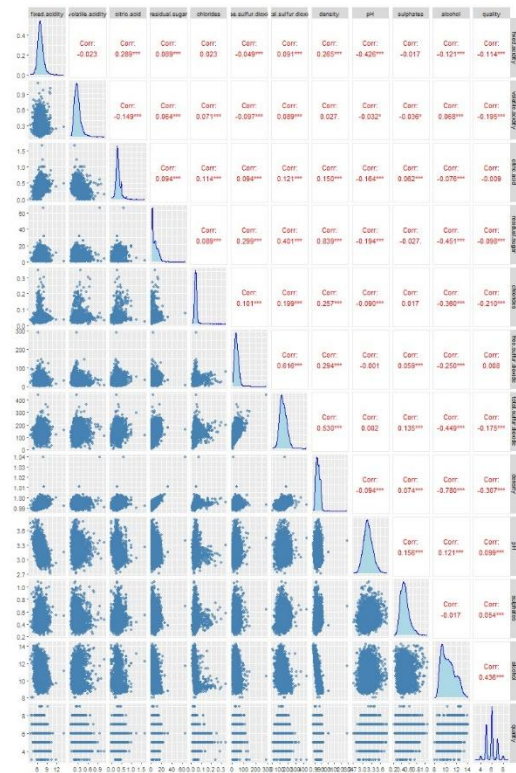
IV. Results and Discussion

4.1 Exploratory Findings

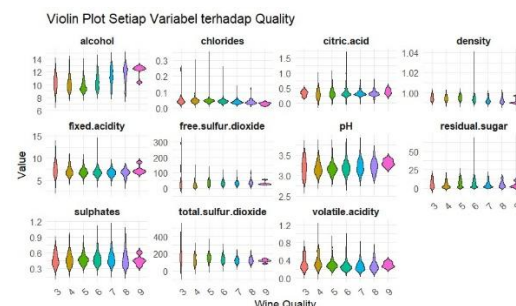
The correlation heatmap provides a clear overview of how the physicochemical attributes relate to one another and to the wine quality. Several important trends can be observed.

First, **alcohol** shows the strongest positive correlation with wine quality, which is also visually apparent in the heatmap through its deeper red shading toward the “quality” axis. This confirms that higher alcohol levels are generally associated with higher-rated wines. In contrast, **density** exhibits a stronger negative correlation with both alcohol and quality, represented by darker blue tones. This pattern suggests that wines with lower density which often indicates higher

alcohol and lower residual sugar tend to receive better quality scores.



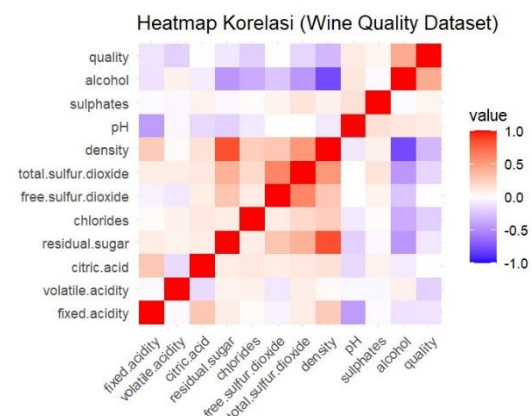
Other features such as **volatile acidity** and **chlorides** show moderate negative correlations with quality. Their blue gradients within the heatmap reflect this tendency, and align with the understanding that excessive acidity or salinity can negatively affect sensory perception. Meanwhile, **residual sugar**, **free sulfur dioxide**, and **total sulfur dioxide** display lighter, more neutral colors, indicating weaker or more variable correlations. These attributes may still influence wine characteristics but do not strongly differentiate quality classes on their own.



Variables like **pH**, **sulphates**, and **citric acid** show only minimal correlation with quality,

which is visible in the near-white blocks along the quality row and column. This suggests that their influence is either indirect or overshadowed by stronger predictors such as alcohol and density.

Overall, the heatmap supports the conclusion that wine quality in this dataset is primarily shaped by a combination of alcohol content, density, and acidity-related properties. These exploratory insights provide a foundation for understanding why certain variables emerge as key predictors in the subsequent machine learning model.



4.2 Model Performance

The Random Forest classifier was trained using an 80/20 train-test split and achieved an accuracy of **69%** on the testing dataset. This performance is relatively strong given the imbalanced distribution of quality classes, where most samples fall into mid-range categories (quality 5 and 6). As a result, the model tends to predict these classes more accurately compared to the extreme quality scores.

	Actual						
Predicted	3	4	5	6	7	8	9
3	0	0	0	0	0	0	0
4	0	6	2	0	0	0	0
5	3	9	204	56	4	0	0
6	3	9	86	364	76	17	0
7	0	1	2	23	92	13	0
8	0	0	0	0	0	10	0
9	0	0	0	0	0	0	0

Accuracy: 0.69

The confusion matrix showed that the model performs best in predicting middle-range quality classes (such as 5 and 6), which also represent the majority of samples in the dataset. In contrast, classes at the extreme ends of the quality spectrum were more prone to misclassification, largely due to their smaller sample sizes and less distinct features.

	precision	recall	F1
3	NaN	0.000000	NaN
4	0.750000	0.240000	0.363636
5	0.7391304	0.6938776	0.7157895
6	0.6558559	0.8216704	0.7294589
7	0.7022901	0.5348837	0.6072607
8	1.0000000	0.2500000	0.4000000
9	NaN	NaN	NaN

Precision, recall, and F1-scores also vary across classes, with lower performance observed in classes that have fewer data points or insufficiently distinctive chemical profiles. These variations underline the complexity of modeling wine quality purely from physicochemical attributes.

4.3 Variable Importance

Feature importance analysis from the Random Forest model reveals several key predictors. Based on computed importance values, the most influential variables are:

1. **Alcohol**
2. **Volatile acidity**
3. **Density**
4. **Free sulfur dioxide**
5. **Total sulfur dioxide**

Alcohol stands out as the strongest predictor of wine quality, aligning with both literature and exploratory findings. Density and volatile acidity follow closely, highlighting their significant roles in shaping wine character and quality perception. Residual sugar, pH, and chlorides fall into the mid-range of importance, whereas citric acid and sulphates contribute the least to the model though

they may still participate in interaction effects captured by the ensemble approach.

Taken together, these findings confirm that wine quality is influenced by multiple interacting variables rather than a single dominant attribute. The Random Forest model effectively captures these nonlinear relationships, making it a suitable predictive method for this dataset.

V. Conclusion

This study demonstrates that the Random Forest classifier can effectively predict white wine quality using physicochemical attributes. Exploratory analysis showed that alcohol, density, and volatile acidity have the strongest relationships with wine quality, while features such as pH, sulphates, and citric acid contribute less. With an accuracy of 69%, the model performs well despite class imbalance and overlapping feature distributions, particularly excelling in mid-range quality classes as reflected in the precision, recall, and F1-scores.

Feature importance analysis confirmed alcohol, density, and volatile acidity as the key predictors, aligning with established domain knowledge and supporting the model's interpretability. Overall, the findings indicate that physicochemical measurements can provide reliable, objective predictions of wine quality. Future improvements may involve addressing class imbalance, testing alternative models such as Gradient Boosting or SVM, and incorporating additional sensory or categorical data to enhance performance.